

**Syllabus
Of
Probability and Statistics
Course Code: STAT-2311
Course Title: Probability and Statistics
Credit Hours: 2
Contact Hours: 2 per week**

Section-A (Mid-term: 30 Marks)

1. Preliminary idea of Statistics: Origin, History and Development Statistics, Definition of Statistics, Characteristics, Function, Limitations, Necessity & importance of Statistics, The role of Statistics in Engineering, Population and Sample, Variable and Constants, Different types of variables, Parameter, Statistic, Scale of measurement, Statistical data, collecting engineering data, Preparation of Questionnaire and Schedule, Presentation and Classification of Data, Construction of Frequency distribution, Graphical presentation of Frequency distribution.

2. Measures of Central Tendency: Different types of mean with their properties and relationship, Quintiles with their graphical presentation, Application of different measures of central tendency.

3. Measures of Dispersion of a Series of Data: Range, Standard Deviation, Mean Deviation, Quartile Deviation, Variance and Standard Deviation, Coefficient of Variation and their uses, Properties and applications of different measures of dispersion, moments, skewness and kurtosis.

Section-B (Final Exam: 50 Marks)

Group-A (20 Marks)

4. Correlation Theory and Regression Analysis: Simple Correlation and its measure, Scatter Diagram, properties of simple correlation coefficient, Spearman's Rank correlation coefficient, Simple linear regression, properties of regression coefficient, regression curve, regression equation, Least-square method of curve fittings, Co-efficient of determination, Theorems & Problems.

5. Basic concepts of probability: History, meaning and Scope of probability, Approaches of defining probability: Classical, Empirical, Subjective and Axiomatic probability, Experiment, random experiment, Sample Space, Event, different types of events, Tree diagram, Conditional probability and independence, some elementary theorems on probability, and conditional probability, Laws of Probability – Additive and Multiplicative Law, Prior probability, Posterior probability, total probabilities & Bayes' theorem and their applications.

Group-B (30 Marks)

6. Random Variables and Mathematical Expectation: Discrete and continuous random variables, Probability mass function and density function, Distribution function and their properties, Mathematical expectation and variance of a random variables. Theorems & Problems.

7. Probability Distributions: Binomial distribution, Poisson distribution and Normal distribution – Their properties, uses, Theorems & Problems.

8. Test of hypothesis and Estimation: Preliminary ideas of statistical hypothesis test, level of significance, one tailed and two tailed test, p-value, Test of significance, Test regarding single mean, test of equality of two mean, test of equality of several means (Analysis of variance), test regarding proportion. General concepts of Estimation.

Text Books:

1. Montgomery Douglas C. and Runger George C. Applied Statistics and Probability for Engineers "Fifth Edition "John Wiley and Sons Inc." (2011). ISBN: 978-0-470-05304-1)
2. "Walpole R.E. Myers R.H. Myers S.L. & Ye K.Y." Probability and Statistics for Engineers and Scientist. 9th ed. Prentice Hall (2012). ISBN 978-0-321-62911-1
3. R.N. Shill & S.C. Debnath An introduction to the theory of Statistics 4th Edition STAR Publication 2016

Reference Books:

1. "Probability Statistics and Random Processes for Electrical Engineering" 1st Edition Cambridge University Press 2014 ISBN-13: 978-1107439887 Alberto Leon-Garcia
2. "E. Keshava Reddy E.Rukmangadachari" Probability and Statistics 5th Edition Pearson Education India 2015 ISBN: 9789332558229